Aizhan Kozhamuratova

Software Engineer — AI & Machine Learning Systems Developer

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Summary

Highly analytical **Software Engineer** with over 3 years of experience in designing **Scalable Backend Architectures** and implementing **Artificial Intelligence** solutions. Expert in developing **High-Load Microservices** using **FastAPI** and **Django**, with a deep focus on **Asynchronous Programming** and **Database Optimization** via **Asyncpg**. Proven expertise in **Machine Learning** research, specifically in **Few-Shot Learning** and **Computer Vision** using **PyTorch**. Demonstrated success in managing **Large-Scale Data Processing** (100k+ records), automating complex **ETL Pipelines**, and deploying robust **Cloud Infrastructure** within **GCP** and **AWS** environments to drive measurable business growth and technical excellence.

Technical Skills

Languages & Databases: Python, SQL (PostgreSQL, PostGIS), JavaScript, HTML/CSS, Redis, MinIO, NoSQL
Frameworks & Backend: FastAPI, Django REST Framework, AsyncIO, asyncpg, Celery, RabbitMQ, RESTful APIs
AI & Machine Learning: PyTorch, Scikit-learn, Few-Shot Learning, Computer Vision, Pandas, NumPy, Matplotlib
Architecture & Tools: Microservices, DTO/Facade Patterns, Distributed Systems, Event-Driven Architecture
Cloud & DevOps: Docker, Kubernetes, GCP, AWS, CI/CD, Linux (Ubuntu), Git, GitHub Actions
Data & Analytics: Process Mining, Geospatial Data Analysis, ETL Pipelines, Statistical Modeling, MLflow

Experience

Software Engineer (AI & Infrastructure) — Aug. 2025 – Present
Alatau City Garant — *Almaty, Kazakhstan*

- Lead the backend development of a high-load **Enterprise Resource Planning** ecosystem, managing end-to-end delivery of **Distributed Microservices** and AI modules.
- Engineered a robust **Event-Driven Architecture** using **FastAPI** and **RabbitMQ**, achieving a 40% increase in message processing throughput for real-time tasking.
- Implemented advanced **Concurrency Control** mechanisms with **AsyncIO** and **SQLAlchemy 2.0**, optimizing connection pooling for 1,000+ simultaneous database sessions.
- Architected a **Scalable Data Layer** utilizing **PostgreSQL Partitioning** and **Materialized Views**, reducing complex reporting query latency by over 65% for 100k+ records.
- Developed custom **Middleware Components** for centralized **OAuth2/JWT Authentication** and request logging, enhancing system security and auditability across all services.
- Designed a **Fault-Tolerant Integration** engine with **1C Soap/Rest APIs**, employing **Idempotency Keys** to ensure zero data loss during financial record synchronization.
- Automated deployment workflows via **Docker Compose** and **GitLab CI/CD**, reducing the **Mean Time to Recover (MTTR)** by 50% through containerized health checks.

AI Research Software Engineer — Jan. 2025 – Mar. 2025
Optimodal — *Hof, Germany*

- Spearheaded the development of a **Geospatial Intelligence** platform, focusing on **Big Data Analysis** of demographic trends and urban infrastructure optimization.
- Optimized **Spatial Indexing** using **GiST** and **BRIN** in **PostGIS**, enabling sub-second execution for complex intersection queries on 143k+ geographic polygons.
- Developed **Vectorized Data Pipelines** using **GeoPandas** and **Dask**, facilitating parallel processing of large-scale demographic datasets across multi-core environments.
- Integrated **Statistical Inference** models to identify **Demographic Anomalies**, providing high-resolution population density heatmaps with **H3 Hexagonal Hierarchical** grids.
- Refined **Data Ingestion Workflows** by implementing **ETL Validation** scripts, ensuring 99.9% **Data Consistency** between raw CSV sources and production databases.
- Leveraged **Advanced SQL Analytics** including **Window Functions** and **Recursive CTEs** to model complex population migration patterns over multi-year periods.
- Collaborated on **UI/UX Integration** using **Leaflet.js**, mapping backend spatial outputs to interactive front-end components for real-time decision-making support.

Backend Developer (IoT & Cloud) Dec. 2022 – Mar. 2024
Techno Park *Almaty, Kazakhstan*

- Designed a scalable **Smart Home IoT Platform** backend using **Django REST Framework**, coordinating communication between sensors, users, and cloud controllers.
- Engineered an **Asynchronous Notification Service** with **Celery** and **Redis**, handling 10k+ daily push alerts for sensor triggers with sub-200ms latency.
- Architected **Real-Time Data Streaming** via **WebSockets** to provide users with instantaneous device status updates and live telemetry monitoring across the dashboard.
- Implemented **Multi-Tenant Database** structures, ensuring strict **Data Isolation** and security for individual user accounts and their associated IoT device networks.
- Developed **Firebase Cloud Messaging** integrations for mobile synchronization, optimizing **Battery Consumption** on client devices through efficient payload delivery.
- Optimized **API Documentation** using **Swagger/OpenAPI**, streamlining frontend-backend integration and reducing developer onboarding time by 20%.
- Deployed **Monitoring & Alerting** stacks with **Sentry**, identifying and resolving critical production bottlenecks before they impacted the end-user experience.

Research & Publications

Deep Learning Research Project Apr. 2025 – May 2025
Kazakh-British Technical University *Almaty, Kazakhstan*

- Developed and benchmarked state-of-the-art **Few-Shot Learning** architectures including **Prototypical Networks** and **MAML** for high-precision medical imaging classification.
- Achieved a peak accuracy of **89.07%** on MRI datasets by implementing **Metric-Based Learning** techniques, outperforming traditional supervised models under limited label constraints.
- Built scalable **Episodic Training** pipelines in **PyTorch**, processing 3,000+ MRI images with customized **Image Augmentation** to enhance model generalization and robustness.

Education

HOF University Hof, Germany
BS in Computer Science (Exchange) *Sep. 2024 – Dec. 2025*

- **Relevant Coursework:** Applied AI, Applied Robotics, Data Science, Advanced Software Engineering, Process Mining.

SDU University Almaty, Kazakhstan
BS in Computer Science *Sep. 2021 – June 2026*

- **Relevant Coursework:** Algorithms and Data Structures, Web Development, Computer Systems, Linear Algebra.